

1 Solution — GIAM 3.2

Problem 4

1.1 Problem

Prove that for all integers a, b, c : if $a \mid b$ and $a \mid (b + c)$, then $a \mid c$.

1.2 Proof

***Proof.** Suppose $a \mid b$ and $a \mid (b + c)$. By the definition of divides there are integers k and m with*

$$b = ak \text{ and } b + c = am.$$

Since $c = (b + c) - b$,

$$c = am - ak = a(m - k).$$

Now $m - k$ is an integer, since $\mathbb{Z}$ is closed under subtraction. So $c = a \cdot (m - k)$ with $m - k \in \mathbb{Z}$, which is precisely $a \mid c$. ■

1.3 The Idea

The conclusion is about c , so write c using only the two things you were given.

$$c = (b + c) - b$$

The proof

$$a \mid b \Rightarrow b = ak, k \in \mathbb{Z}$$

$$a \mid (b+c) \Rightarrow b + c = am, m \in \mathbb{Z}$$

$$c = (b + c) - b = am - ak = a(m - k)$$

$m - k \in \mathbb{Z}$ (closure of $\mathbb{Z}$ under subtraction)

so $a \mid c$ ■

A worked instance

a	$b = ak$	$b + c = am$	$c = a(m - k)$	check
3	$12 = 3 \cdot 4$	$30 = 3 \cdot 10$	$3 \cdot (10 - 4) = 18$	✓
5	$-15 = 5 \cdot (-3)$	$20 = 5 \cdot 4$	$5 \cdot (4 + 3) = 35$	✓
7	$7 = 7 \cdot 1$	$-42 = 7 \cdot (-6)$	$7 \cdot (-6 - 1) = -49$	✓

The general fact behind it

if $a \mid x$ and $a \mid y$, then $a \mid (sx + ty)$ for all integers s, t
 a divides any integer linear combination of things it divides

this problem is the case $x = b+c, y = b, s = 1, t = -1$

Equivalently: the multiples of a are closed under addition and subtraction — knowing two of $b, c, b+c$ pins down the third.

Figure 1.1: Top: a highlighted box giving the key rearrangement, c equals $(b$ plus $c)$ minus b . Below, the proof: a divides b gives b equal to ak with k an integer; a divides b plus c gives b plus c equal to am with m an integer; then c equals $(b$ plus $c)$ minus b equals am minus ak equals a times $(m$ minus $k)$; and m minus k is an integer by closure of the integers under subtraction, so a divides c . A table of worked instances: for a equals 3, b equals 12 which is 3 times 4, b plus c equals 30 which is 3 times 10, so c equals 3 times $(10$ minus $4)$ which is 18. For a equals 5, b equals minus 15 which is 5 times minus 3, b plus c equals 20 which is 5 times 4, so c equals 5 times 7 which is 35. For a equals 7, b equals 7, b plus c equals minus 42, so c equals 7 times minus 7 which is minus 49. A green panel gives the general fact: if a divides x and a divides y , then a divides sx plus ty for all integers s and t — a divides any integer linear combination of things it divides — and this problem is the case x equals b plus c , y equals b , s equals 1, t equals minus 1. A closing panel: equivalently, the multiples of a are closed under addition and subtraction, so knowing two of b, c, b plus c pins down the third.

The conclusion is a statement **about** c , so — as in Problem 3.1.8 — the move is to isolate c using only what was given. Both hypotheses concern b and $b + c$, and the identity

$$c = (b + c) - b$$

expresses the target as a difference of exactly those two quantities. Everything else is substitution and closure. Verified exhaustively for nonzero a and $|b|, |c| \leq 40$.

1.4 Remark — The General Fact: Divisors Respect Linear Combinations

This theorem is one instance of a much more useful lemma:

$$a \mid x \text{ and } a \mid y \implies a \mid (sx + ty) \quad \text{for all integers } s, t.$$

The proof is the same one line: $x = au$, $y = av$ give $sx + ty = a(su + tv)$, and $su + tv \in \mathbb{Z}$. (Verified on 5000 random cases.)

The present problem is the special case $x = b + c$, $y = b$, $s = 1$, $t = -1$, since $1 \cdot (b + c) + (-1) \cdot b = c$. Recognising a problem as an instance of a general lemma is usually worth more than the problem itself — the lemma also gives, for free, that $a \mid (b + c)$, $a \mid (b - c)$, $a \mid 5b - 3c$, and so on.

1.5 Remark — Any Two of the Three Determine the Third

The three quantities b , c , and $b + c$ satisfy one linear relation, so **knowing a divides any two of them forces it to divide the third**:

given	deduce	via
$a \mid b, a \mid c$	$a \mid (b + c)$	$b + c = b + c$
$a \mid b, a \mid (b + c)$	$a \mid c$	$c = (b + c) - b$ — this problem
$a \mid c, a \mid (b + c)$	$a \mid b$	$b = (b + c) - c$

Table 1.1

All three are the same lemma with different coefficients, and the symmetry is a good check that nothing special is happening in the stated version.

1.6 Remark — Multiples of a Form a Closed System

Restated set-theoretically: the multiples of a ,

$$a\mathbb{Z} = \{\dots, -2a, -a, 0, a, 2a, \dots\},$$

are **closed under addition and subtraction**. Adding or subtracting two multiples of a always gives another multiple of a . So the theorem says: if b and $b + c$ both lie in $a\mathbb{Z}$, then so does their difference c .

In ring-theoretic language $a\mathbb{Z}$ is an **ideal** of $\mathbb{Z}$, and the linear-combination lemma above is exactly the defining property of one. This is also the structure behind Bézout's identity and the Euclidean algorithm of Section 1.5: the algorithm works precisely because replacing a pair by a remainder — an integer combination — never leaves the set of common multiples.

1.7 Remark — No Hypothesis on a Is Needed

Unlike Problem 3.1.13, this theorem needs **no positivity assumption**, and it even holds when $a = 0$. If $a = 0$ then $0 \mid b$ forces $b = 0$, and $0 \mid (b + c)$ forces $b + c = 0$, hence $c = 0$ and indeed $0 \mid c$. The proof handles this without comment because it never divides by a — it only ever *multiplies*.

That is worth noticing. Proofs that manipulate b/a need $a \neq 0$; proofs that stay in the form $b = ak$ do not. Keeping the definition in its multiplicative form is what makes the argument work uniformly.

1.8 Remark — Contrast With Transitivity

Problem 3.1.11 proved $a \mid b$ and $b \mid c$ imply $a \mid c$; this one proves $a \mid b$ and $a \mid (b + c)$ imply $a \mid c$. They look similar but use different structure:

- **Transitivity** chains two divisibilities and merges the multipliers by **multiplication**: $c = a(km)$.
- **This theorem** combines two divisibilities by the *same* divisor and merges the multipliers by **subtraction**: $c = a(m - k)$.

The first exploits associativity of $\times$; the second exploits distributivity of $\times$ over $-$. Both then invoke closure of $\mathbb{Z}$ to certify the new multiplier. Seeing which algebraic law a proof actually rests on is a good habit — it tells you immediately how far the result will generalise.